

■ B.7.3 Files

When you use a command like `<<name`, *Mathematica* uses the following procedure to try and resolve the name of the file you want:

- If the name starts with `!`, *Mathematica* treats the remainder of the name as an external command, and uses a pipe to this command.
- *Mathematica* scans the name for the characters `*`, `$`, `~`, `?`, `[`, `"`, `\`, ``` and `'`. If it finds any of these characters, then it passes the complete name to your operating system or shell. Under `UNIX`, *Mathematica* asks the shell to `echo` the file name, so that whatever globbing, quote-stripping or variable expansion the shell usually does is done. So long as the resulting file is unique, *Mathematica* then uses its name.
- If the file you specify is to be used for output, then *Mathematica* performs no further transformations.
- If the file you specify is to be used for input, then *Mathematica* can search a list of directories to try and find the file. It will not do this if the name you gave contained any of the special characters listed above, or if it starts with a slash. Otherwise, *Mathematica* goes down the list `$Path`, taking each element as a string which specifies a directory name. *Mathematica* concatenates each directory name, followed by a slash, with the name you specified, and then looks for a file with the resulting name.

<code>!command</code>	a pipe to an external command
<code>/directory/name</code>	a file in an absolutely-specified directory
<code>\$DIR/name</code> etc.	a file whose name is to be resolved using the shell
<code>name</code>	a file to be searched for in the list of directories <code>\$Path</code>

Forms of file names.

When *Mathematica* starts up, the initial value of `$Path` is usually set by a `-run` command line argument in the `math` shell script. `$Path` typically contains the current directory, the user's home directory, the main *Mathematica* directory and the directory or directories for *Mathematica* notebooks and packages.

Note that the `$Path` mechanism is used *only* for files being read into *Mathematica*. Names of files to be written out by *Mathematica* are always specified relative to the current directory.

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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<code>"stdout"</code>	standard output
<code>"stderr"</code>	standard error

Special files recognized by *Mathematica*.

If *Mathematica* runs into a syntax error in an input file, it reports the error in the standard form *filename: linenumber: syntax error in inputstring*. By dissecting this form, you can make external programs, such as text editors, go to lines with syntax errors.